VAISHNAVI PUJALA

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SUMMARY

Associate Product Engineer with an M.S. in CS who published two Springer papers and shipped **agentic AI** and **LLM** applications, by building **RAG** pipelines, **federated learning** systems, and full-stack services with **Python**, **FastAPI**, and **React**

EXPERIENCE

Aptean, Associate Product Engineer Oct 2026 – Present

- Contributed to **AI-driven product features** by building and integrating **MCP servers** and **LLM** tooling with senior engineers and **product management** in **Agile sprints**
- Supported release quality by testing and troubleshooting **MCP server** integrations, writing **requirements documentation**, and validating changes through **code reviews**

NeuroHeart AI Labs LLC, AI Developer Intern Jun 2026 – Oct 2026

- **QA-tested** and hardened a **conversational voice AI ordering agent** for a live restaurant client, uncovering and documenting critical production bugs (hallucinated entities, multi-attribute order failures) and iterating **LLM prompts** for a **RAG**-based phone agent, improving **eval accuracy from 60% to 80%** across a structured test suite
- Shipped a **Next.js** catering **dashboard** end-to-end and drove **compliance and competitive research** (TCPA, call-recording law) that directly informed the team's **voice platform** (Deepgram vs. ElevenLabs) decision, reducing vendor evaluation time by **50%** ahead of production rollout

Georgia State University, Graduate Research Assistant Aug 2025 – May 2026

- Delivered a **70.48%** cross-domain accuracy an **8.7%** improvement over baseline by designing a **privacy-preserving federated HAR system** (**WISDM dataset**) using **CNN-BiLSTM** with **Temporal Self-Attention** and **DANN**
- Benchmarked **5 domain adaptation strategies** (Source-Only, Centralized DANN, **Federated DANN via FedProx**, Attention-augmented DANN, Personalized Federated DANN) using **Gradient Reversal Layer** scheduling and **per-client personalized heads**, maximizing target-domain generalization while keeping all raw sensor data **on-device** for privacy

Georgia State University, Graduate Teaching and Lab Assistant Aug 2024 – May 2025

- Improved learning outcomes for **100+** undergraduate students in CSC3320 **System-Level Programming**, by designing and delivering hands-on labs on **C, C++, memory management, pointers**, and **Python**-based **data visualization**
- Refined course assessment strategies by analyzing student performance data and co-developing **regression/classification** lab exercises with faculty, strengthening the course's data analytics curriculum

Atal Incubation Centre, AIC-GNITS Foundation, Computer Vision & ML Engineer Jul 2023 – Jul 2024

- Achieved **90%** detection accuracy on copy-move image forgery detection, by constructing and optimizing a **computer vision** pipeline using **CNN** and **VGG-16** for preprocessing, feature extraction, and model training in **Python, TensorFlow/Keras, OpenCV**
- Powered real-time image upload and forgery localization for end users, by building a **full-stack ML inference system** integrating a **Flask** backend with an interactive **HTML/CSS/JavaScript** frontend, published in Springer ICCCE 2024

Innovation and Incubation Cell – GNITS, AI/ML Research Associate Aug 2022 – Jun 2023

- Reached **91% accuracy** on a **heart disease prediction** system, by comparing **Random Forest, Naïve Bayes, Logistic Regression** and applying **feature selection, hyperparameter tuning**, and **cross-validation** published in Springer ICDSMLA 2023
- Surfaced actionable health risk patterns for **AI-driven healthcare analytics** workflows, by performing **exploratory data analysis** and building interactive visualizations with **Plotly, Seaborn, Matplotlib**

PROJECTS

Code Sentinel | Python, FastAPI, LangGraph, LLaMA 3, React, PostgreSQL, Redis, Docker

- Automated vulnerability detection across **GitHub repositories**, by engineering a **multi-agent security scanner** with **LangGraph** orchestrating **4 parallel agents** (SAST, Secrets, Dependency, Fix), powered by **LLaMA 3** fix generation
- Enabled real-time scan visibility and reliable async processing, by building a **Celery + Redis** pipeline with **WebSocket** progress streaming and **JWT**-authenticated **FastAPI** endpoints persisting results to **PostgreSQL**

PUBLICATIONS

- **Springer Journal:** An effective Prediction of Heart Diseases using Machine Learning Techniques, ICDSMLA 2023 – Machine Learning
- **Springer Journal:** An approach to Detect Copy Move Forgery Detection using Deep Learning Techniques, ICCCE 2024 – Computer Vision

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, Go, C, C++, SQL, HTML, CSS
Frameworks & Libraries: React.js, Next.js, Node.js, Angular, FastAPI, Flask, Django, Flutter, Tailwind CSS, PyTorch, TensorFlow, Keras
Cloud & DevOps: AWS (S3, SQS, SNS, Lambda), GCP, Docker, Kubernetes, GitHub Actions, CI/CD, Linux, Bash
Databases: PostgreSQL, MongoDB, DynamoDB, Redis, NoSQL
Tools & Concepts: REST APIs, Microservices, System Design, OOP, Unit Testing, Agile, Kafka, NLP, LLMs, Tableau, Power BI, Data Structures, Secure Coding

EDUCATION

Georgia State University, Atlanta, GA, USA | M.S. Computer Science Aug 2024 – May 2026
Coursework: Machine Learning, Neural Networks, Deep Learning, LLMs, Federated Learning